Measuring the External Field

A cortical-proxy content observable and the mean-field bound on media-driven opinion change

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The gap this project addresses

Statistical physics models media as an **external field** h acting on a population of coupled opinion states.

In that literature the field is **assigned by hand**. A population is shown to tip, without establishing that any real influence is strong enough to tip it.

The two questions that follow

1. Can a field be *measured* from content itself, rather than assumed?
2. How strong would the coupling have to be for a measured spread to matter?

The instrument

Text in, predicted cortical response out. Five stages.

1. **Speech synthesis** — the encoder expects naturalistic audiovisual input.
2. **Word timings** — forced alignment recovers per-word timing.
3. **Embedding** — text and audio through the pretrained backbones.
4. **Prediction** — TRIBE v2 returns 20,484 fsaverage5 cortical vertices per time point.
5. **Reduction** — mean-pooled over time, reduced to two ROI means.

Public and running: `monarch-4iy.pages.dev`

Three things the released model turned out to be

Established by inspecting the artifact, not by reading the paper about it.

1. **Cortical only.** Config declares `mesh: fsaverage5`; output is $(T, 20484) = 2 \times 10242$. There is no subcortical head. **The amygdala is not predicted**, so the proposal's Eq. (4) cannot be computed at all.

2. **Averaged over subjects.** The loader forces `average_subjects = True`. The subject term is unavailable.

3. **Output is standardised.** ROI means sit near zero, so a ratio observable sign-flips or explodes. Undefined for 69 of 400 items, 17.25%.

Each is a property of the instrument, found by testing it.

The observable

Two cortical unions, resolved once into vertex index sets and cached:

affective salience	1,030 vertices
deliberative control	851 vertices

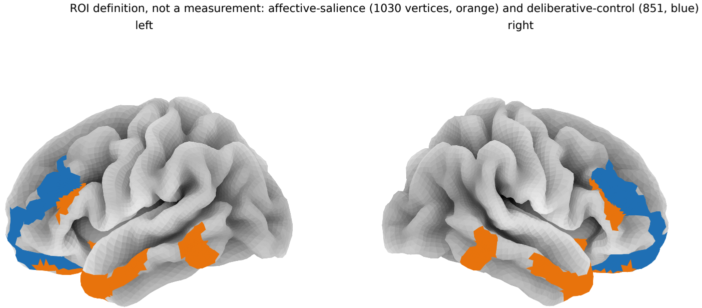
The index is the signed difference, always defined:

$$\mathrm{NAA}_{\mathsf{signed}} = A_{\mathsf{aff}} - A_{\mathsf{del}}$$

Positive means the affective set is predicted to lead.

It is a **cortical proxy**. Not a measurement of any person, and never called the amygdala.

The two regions



The corpus

400 items, four categories of 100, matched on length.

category	source	mean words
fear-activating	ISOT-fake	167.4
high outrage	SemEval-2019 Task 4	163.5
reward hook	Webis-Clickbait-17	163.7
neutral informational	PubMed, ISOT-true	162.2

Standard deviations near 11. Every row verified against the source corpus on text, category and identifier before any analysis ran.

Power computed *before* the analysis: smallest detectable $\eta^2 = 0.0268$, smallest detectable AUC $= 0.5916$.

Result: the categories separate

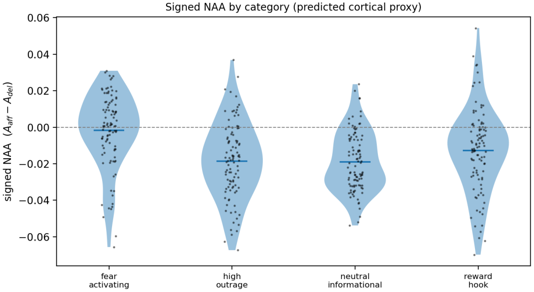
Four-category ANOVA, $n = 400$:

$$\eta^2 = 0.1068$$

$$F = 15.779, \quad p = 1.03 \times 10^{-9}$$

Against a design powered to detect 0.0268. Power at the observed value is 1.0.

Against the pre-scan `manipulative` label: $\mathrm{AUC} = 0.6274$, clearing the detectable 0.5916.



Where the separation comes from, and the confound

Welch tests against the neutral baseline:

category	d	p
fear-activating	$+0.939$	3.3×10^{-10}
reward hook	$+0.319$	0.025
high outrage	$+0.030$	0.832

Outrage does not separate at all — the category the proposal expected to separate most.

Each category is drawn from exactly one source dataset. So the separation is between corpora and cannot be attributed to framing. Length is controlled; provenance is not. This is stated before the result, not in limitations.

One piece of evidence against a pure source artifact: outrage has its own distinct source and does not separate.

Reliability: measured, not assumed

A second GPU session rescanned all 400 items with identical text, code and regions.

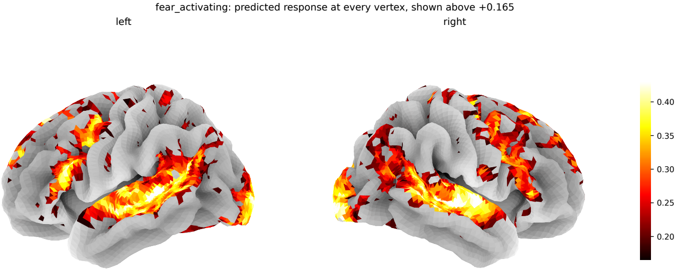
	r	ICC(2,1)	$\mathrm{sd}(\Delta)/\mathrm{sd}_{\text{between}}$
$\mathrm{NAA}_{\text{signed}}$	0.8763	0.8725	0.484
A_{aff}	0.8759	0.8713	0.489
A_{del}	0.8576	0.8550	0.521

The separation replicates: $\eta^2 = 0.0888$ on session two, against 0.1068 on session one. Both clear the detectable floor.

Per-item direction does not. 51 of 400 items, 12.8%, reverse sign between sessions.

Group-level claims are supportable. **No per-item verdict appears anywhere** — not in the analysis, the figures, or the public interface.

What one item's prediction looks like



All 20,484 vertices, thresholded at the map's 70th percentile so the anatomy stays readable.

The physics: mean-field, with coefficients derived

Self-consistency for a coupled population under field h:

$$m = \tanh(\beta J m + h)$$

Expanding $\operatorname{artanh}(m) \approx m + m^3/3$ gives the Landau free energy

$$F(m) = am^2 + bm^4 - hm, \qquad a = \frac{1-\beta J}{2}, \quad b = \frac{1}{12}$$

Coefficients **derived from the self-consistency condition**, not asserted. The proposal carried $a = 1 - \beta J$, $b = (\beta J)^3/3$, which do not follow from it. Corrected.

Solver checked against the exact exponents: $\beta = 1/2$, $\gamma = 1$, $\delta = 3$.

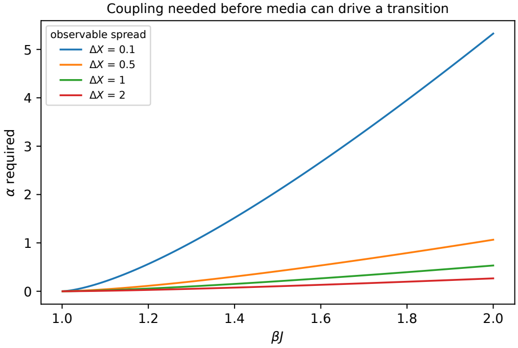
The bound, and why no $\hat{\alpha}$ is quoted

Calibrating α against the corpus **does not identify it**: the fit performs worse than the sample mean out of sample, and the estimate scales with an unmeasured βJ.

So no value is quoted. Anywhere. Not in a table, not in a caption.

What survives is a **necessary condition** on any content observable used as a field:

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}$$



Measured spread $\Delta X = 0.1241$ gives $\alpha \geq 0.169$ at $\beta J = 1.10$, 1.655 at 1.50, and 4.294 at 2.00.

Paper 3: does the encoder track real cortex?

A commercial lab reports the released checkpoint, in the averaged configuration this project runs, is **anti-correlated** with real cortex. Four subjects, one stimulus, not independently replicated.

Measured here, from public data: inter-subject noise ceiling $= 0.1517$, CI $[0.1484, 0.1551]$, four subjects, 1000 of 1000 Schaefer parcels, episode `s01e01a`.

A first value of 0.2247 **was withdrawn.** Two silent defects: an episode selected by position rather than name, and an orientation guard that could not fire. Replaying the old path reproduces the wrong number exactly, which is what proved the cause.

The prediction pass is specified in the paper and is **on its final run**.

What is claimed, and what is not

Claimed

- A content observable that can be measured, with its reliability stated.
- That it separates these four corpora at $\eta^2 = 0.1068$, power attached.
- A bound any candidate field observable must satisfy, applicable before data is collected.

Not claimed

- That the index detects manipulation. Category is confounded with source.
- Anything about the amygdala. The checkpoint does not predict it.
- Any value of the coupling α. The corpus does not identify it.
- Any verdict on an individual item. 12.8% reverse between sessions.

Output

- **Dissertation**, 72 pages. Complete draft.
- **Paper 1** — the bound and the screening criterion it yields. 12 pages, for *Physica A*. Independent of the scan.
- **Paper 2** — the instrument, the 400-item corpus, the field bound. 12 pages, for *Physica A*.
- **Paper 3** — the measured noise ceiling and the replication. 6 pages, for *Imaging Neuroscience*. One computation on its final run.

Every number in all four comes from a re-runnable script. The instrument, the corpus and the analysis are public at `github.com/brn-mwai/monarch`.

Thank you

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monarch-4iy.pages.dev